

UAV Emergency Rescue — Team Publication Strategy

Turning Objectives 1–5 into publishable papers · prepared by Student 5 (BeiDou Short Message)

The principle: the base method is NOT the contribution. We all use established tools (RTKLIB, PX4/QGC, vision detection, RRT*). A Q1 reviewer rejects “we applied known method X.” Our publishable angle is the **disaster-rescue gap the standard method does not cover**. Find that gap.

Per-objective publishable angles

Obj (owner)	Publishable argument — the disaster-rescue gap	Realistic
1 RTK positioning	RTK reliability under compound-disaster GNSS degradation + a viability-gating scheme for safe autonomous landing (the Level-3 total-failure scenarios ARE the novelty — not RTKLIB)	Q1/Q2
2 QGC / mission	Autonomous precision-landing decision framework gated on real-time RTK viability (land / hold / re-converge / abort)	Q2/Q3 or joint
3 Target detection	Detection-TO-geolocation pipeline (aerial detection → rescue coordinate), OR a disaster-survivor dataset, OR detection under rubble/occlusion/aerial viewpoint	Q2/Q3
4 Path planning	Uncertainty-aware planning to a survivor REGION defined by the rescue search-radius R — not a point (RRT* alone will not publish)	Q2/Q3
5 BeiDou SMC	Field-measured BDS-SMC rescue link + complete 112-bit payload (Paper 1); real-hardware GCS + operations layer (Paper 2)	Q1 / Q1-Q2

Cross-objective papers — often stronger than the parts

Combination	Why it is novel
Obj1 RTK viability → Obj2 landing	Position-confidence-gated autonomous landing (closed safety loop)
Obj5 uncertainty R → Obj4 planning	Uncertainty-aware planning to a survivor region, not a point
Obj3 detect → Obj5 coord → Obj4 path	Detection-confirmed rescue retargeting

Recommended strategy: NOT five separate papers. Target **1–2 strong solo papers** (Obj 5, and Obj 1) + **ONE joint flagship system paper** where every member is an author. Quality over a forced paper each. A few publishable papers beat many unpublishable ones.

Realistic Q1: Obj 5 (Paper 1) and the joint system paper if we manage one real flight. The rest are credible Q2/Q3 or strengthen the joint paper.

One real flight (one drone to one real BeiDou-transmitted coordinate) is the single highest-value addition — it converts the joint paper from 'simulation-validated' to 'demonstrated system' and unlocks better venues.